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Corneal Biomechanics as a Causal Factor in Myopia and Astigmatism: Evidence from Mendelian Randomization

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Purpose: The causal relationship between refractive errors and corneal biomechanical properties remains uncertain. This study aimed to clarify this relationship using Mendelian randomization (MR), offering new insights into the prevention and treatment of refractive errors.

Design: A bidirectional, 2-sample MR analysis.

Participants: Corneal biomechanical data were obtained from 97 653 European participants in the UK Biobank, whereas refractive error data were sourced from the UK Biobank and FinnGen consortia.

Methods: The exposures in this study were identified as corneal biomechanical parameters, specifically corneal hysteresis (CH) and the corneal resistance factor (CRF). The outcomes were defined as refractive errors, including myopia, hyperopia, and astigmatism, along with refractive power, encompassing both spherical and cylindrical power. A meta-analysis was performed to combine the MR estimates from both UK Biobank and FinnGen consortia, with heterogeneity assessed using the Q test and I^2 statistics. Additionally, a reverse MR analysis was conducted to examine the potential causal effect of the refractive status on corneal biomechanics.

Main Outcome Measures: Corneal hysteresis and CRF as causal factors in myopia and astigmatism.

Results: Data from UK Biobank revealed that CH and CRF were protective against the development of myopia (CH: odds ratio (OR) = 0.9936, $P = 7.79 \times 10^{-4}$; CRF: OR = 0.9946, $P = 2.41 \times 10^{-3}$) and astigmatism (CH: OR = 0.9975, $P = 0.02$; CRF: OR = 0.9977, $P = 0.017$). Conversely, increased corneal-compensated intraocular pressure was a risk factor for myopia development (OR = 1.0091, $P = 2.07 \times 10^{-2}$). The meta-analysis, which combined data from both sources, supported a causal relationship between CH and CRF and the development of myopia, although no significant causal link was found for hyperopia. Reverse MR analysis demonstrated a causal effect of spherical power on CH (OR = 1.0664, $P = 4.32 \times 10^{-5}$).

Conclusions: Corneal biomechanical parameters, particularly CH and CRF, may serve as early biomarkers for predicting myopia. This protective role supports their use in clinical screening to enhance early intervention strategies. Corneal-compensated intraocular pressure is a risk factor for myopia and represents a novel therapeutic target. Future research should clarify the underlying mechanisms and assess biomechanical interventions, potentially transforming refractive error management and reducing visual impairment.

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Uncorrected refractive errors, including myopia, hyperopia, and astigmatism, are common ocular conditions that can lead to visual impairment. Myopia, particularly high myopia, significantly increases the risk of pathological ocular changes and profoundly impacts the quality of life of affected individuals.[1,2] In recent years, the incidence of myopia has increased alarmingly, with projections indicating that nearly half of the world's population may be affected by myopia by 2050.[3] Despite the extensive research on refractive errors, the underlying etiologies remain multifaceted and not fully understood. Therefore, elucidating the mechanisms that contribute to the development of refractive errors and their potential biological determinants continues to be a central focus in contemporary ophthalmic research.

The cornea is a critical component of the ocular refractive system and exhibits optical and biomechanical characteristics. Recent studies have found a significant association between corneal biomechanical properties and myopia, underscoring their potential role in refractive error development.[4,5] In vivo studies have indicated that these biomechanical properties are intricately linked to corneal

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morphology and refractive status.[4,6,7] Corneal biomechanics are essential for maintaining its shape.[8] Individuals with high myopia demonstrate significantly reduced corneal biomechanical properties compared to those with moderate-to-low myopia.[9] A meta-analysis demonstrated that corneal biomechanical parameters are significantly greater in the low myopic group, with a mean difference in corneal hysteresis (CH) of 0.73 mmHg and a mean difference in corneal resistance factor (CRF) of 0.20 mmHg.[9] These observations imply that corneal biomechanics undergo substantial alterations as myopia progresses, with a general trend toward weakened corneal biomechanics in advanced myopia.[10–12] However, despite accumulating evidence suggesting that myopia adversely affects corneal biomechanics, these findings remain inconsistent across various studies.[13,14] Consequently, the causal relationship between corneal biomechanical properties and refractive errors remains unclear and warrants further investigation.

Current research predominantly relies on traditional statistical techniques, such as linear regression, which assume a linear relationship between corneal biomechanics and refractive errors. This assumption may limit the comprehensive understanding of corneal biomechanical changes associated with the refractive state. Moreover, these traditional statistical techniques are often vulnerable to confounding variables, complicating the efforts to derive accurate conclusions regarding causality.[15] Mendelian randomization (MR) offers a robust alternative to causal inference by using genetic variants as instrumental variables (IVs) to mitigate confounding and reverse causation biases.[16] This approach has gained significant traction in recent epidemiological studies and offers novel insights into causal inferences.

This study aimed to employ MR to investigate the causal relationship between corneal biomechanical properties and refractive errors. Using data from a large-scale cohort of healthy adults, this study analyzed the patterns of corneal biomechanical changes across various refractive errors and explored their potential etiological roles. The results of this study clarified the complex interplay between corneal biomechanics and refractive errors, offering new scientific evidence for the prevention and management of these conditions.

Methods

Study Design

This study employed a bidirectional 2-sample MR approach to investigate the causal relationship between corneal biomechanical parameters and refractive status. The corneal biomechanical parameters include CH, CRF, corneal-compensated intraocular pressure (IOPcc), and Goldmann-correlated intraocular pressure (IOPg). Specifically, CH, which represents the response of ocular tissues to transient pressure and release from the tonometer, reflects the corneal viscous resistance (the ability to absorb or dissipate energy). Corneal resistance factor represents the cumulative effect of corneal viscous resistance and elastic resistance, reflecting the overall stiffness of the cornea (the ability to resist external forces as a whole). Corneal-compensated intraocular pressure is an intraocular pressure value that is measured independently of corneal characteristics. Goldmann-correlated intraocular pressure is an intraocular pressure value similar to that obtained with a conventional Goldmann applanation tonometer, after correcting the value for the linear correction factor between the results of the Goldmann applanation tonometer and the applanation airflow pressure. The refractive status was defined as the presence of hyperopia, myopia, astigmatism, and spherical or cylindrical power.

Bidirectional MR analysis explores the effects of corneal biomechanical characteristics on refractive errors and assesses the influence of refractive errors on corneal biomechanical parameters. To ensure the validity of causal inference in MR analysis, the selected IVs should meet 3 fundamental assumptions: (1) relevance, which stipulates that genetic variation is strongly associated with the exposure phenotype; (2) independence, which requires that the IVs are not correlated with confounding factors; and (3) exclusion restriction, which states that IVs influence the outcome solely through the exposure factor. An overview of the study design is shown in Figure 1. The study was conducted in accordance with the principles of the Declaration of Helsinki. As all the genome-wide association study (GWAS) data employed in this study were derived from publicly accessible datasets, no further informed consent or medical ethics approval was required.

Corneal Biomechanical Parameter Data

Corneal biomechanical parameters were obtained using the ocular response analyzer (ORA) (Reichert Corp.), and these parameters are currently the only in vivo corneal biomechanical data available in the GWAS database. The ORA employs a noncontact air jet tonometer with a high-speed programmable air jet system and an infrared emission detection system. It quantifies corneal biomechanical parameters by analyzing the changes in infrared reflective intensity on the corneal surface. Upon application of an air pulse, the cornea undergoes indentation and subsequent recovery, experiencing 2 flattening states associated with the corresponding pressures P_1 and P_2. From these pressures, 4 key parameters are derived as follows: (1) $CH = P_1 - P_2$, (2) $CRF = P_1 - 0.7P_2$, (3) $IOPcc = P_2 - 0.43P_1$, and (4) $IOPg = (P_1 + P_2)/2$. Because the parameters were highly correlated between the 2 eyes, we chose the right eye's GWAS data for CH, CRF, IOPcc, and IOPg to maintain consistency. The corneal biomechanical data used in this study were sourced from the UK Biobank consortium and comprised samples from 97 653 individuals of European ancestry. The GWAS utilized 9 851 867 single nucleotide polymorphisms (SNPs).

Refractive Error Data Sources

This study employed data on refractive errors, including myopia, hyperopia, and astigmatism, obtained from the UK Biobank and FinnGen consortia. Further details are provided in Table 1. In the UK Biobank dataset, myopia, hyperopia, and astigmatism were ascertained by the questionnaire item: "Why were you prescribed glasses/contacts?" The FinnGen database describes myopia as "A refractive error in which rays of light entering the eye parallel to the optic axis are brought to a focus in front of the retina when accommodation is relaxed." For astigmatism, the description is "Unequal curvature of the refractive surfaces of the eye. Thus, a point source of light cannot be brought to a point focus on the retina but is spread over a more or less diffuse area." The UK Biobank consortium comprised a sample size of 460 536 individuals of European ancestry, encompassing 37 362 cases and 423 174 controls for myopia, 19 050 cases and 441 486 controls for hyperopia, and 12 513 cases and 448 023 controls for astigmatism. To further investigate the impact of biomechanical parameters on refractive errors, this study

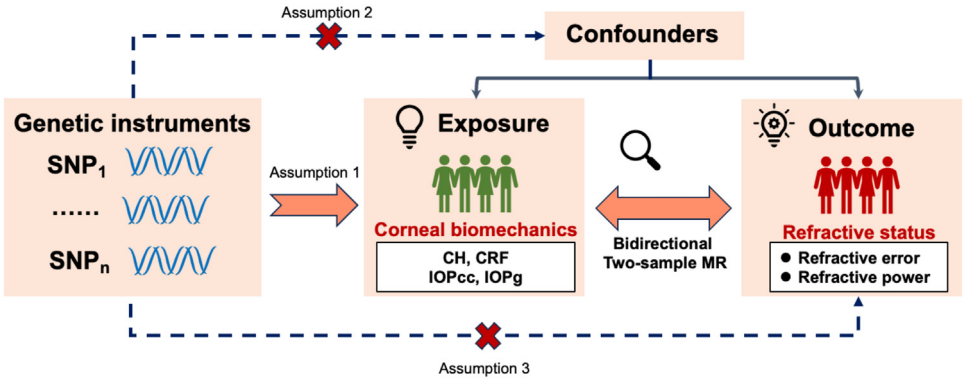


Figure 1. Flowchart of the study design. CH = corneal hysteresis; CRF = corneal resistance factor; IOPcc = corneal-compensated intraocular pressure; IOPg = Goldmann-correlated intraocular pressure; SNPs = single nucleotide polymorphisms.

incorporated data on spherical and cylindrical power from the UK Biobank with a sample size of 99 380.

The FinnGen consortium comprised a sample of 212 571 individuals of European ancestry, of which 1640 were cases and 210 931 were controls. The sample size for hyperopia was 211 717, comprising 786 patients and 210 931 controls. For astigmatism, the sample size was 211 588, comprising 657 patients and 210 931 controls.

Selection of IVs

To select appropriate genetic IVs, we employed rigorous quality control measures. Initially, we conducted a genome-wide screening of SNPs that were significantly associated with exposure, setting the threshold for statistical significance at $P < 5 \times 10^{-8}$. In the event that no SNPs met the aforementioned criterion, the threshold was relaxed to $P < 5 \times 10^{-6}$. Subsequently, to ensure the independence of the genetic variables, linkage disequilibrium was removed from the exposure data using the following criteria: $R^2 < 0.001$ and clump window >10 000 kb. In MR analysis, ambiguous SNPs with nonconcordant alleles and palindromic SNPs were excluded.

To identify any potential for weak instrument bias, we employed the F statistic, calculated according to the following formula: $F = R^2 (n - k - 1)/k (1 - R^2)$. In this context, R represents the proportion of variance in exposure that can be explained using the selected genetic tools. The term n denotes the sample size and k refers to the number of selected IVs. An average F statistic exceeding 10 suggests that the selected IVs are appropriate for this purpose. Furthermore, candidate IVs that did not meet the linkage disequilibrium and Outlier (Mendelian randomization pleiotropy residual sum and outlier) test criteria ($P < 0.05$) were also excluded. The same IV selection procedure was employed for reverse MR analysis.

Mendelian Randomization Analysis

This study employed 2-sample MR analyses conducted using the R software (version 4.4.1) and the TwoSampleMR package (version 0.6.6). In the initial analysis, exposure constituted corneal biomechanical parameter–related GWAS data, whereas refractive error data constituted the outcome. Five methods were employed to assess the causal effects and ensure the robustness of the results. The inverse variance weighted (IVW) method was designated as the primary estimation method, with the additional use of the MR-Egger regression, weighted median, simple mode, and weighted mode as auxiliary techniques. The relative risk associated with the exposure factor was estimated using either the odds ratio (OR) for binary variables or beta coefficients (β) for continuous variables, both accompanied by a 95% confidence interval (CI). A significance threshold was set at $P < 0.05$. A heterogeneity analysis was conducted using the IVW and MR-Egger methods via the Q test,

Table 1. Summary Information on the Genome-Wide Association Study (GWAS) Database in This Study

GWAS ID	Trait	Sample Size	Number of SNPs	Consortium
ukb-b-6353	Myopia	460 536	9 851 867	UK Biobank
finn-b-H7_MYOPIA	Myopia	212 571	16 380 455	FinnGen
ukb-b-18189	Hypermetropia	460 536	9 851 867	UK Biobank
finn-b-H7_HYPERMETRO	Hypermetropia	211 717	16 380 455	FinnGen
ukb-b-10937	Astigmatism	460 536	9 851 867	UK Biobank
finn-b-H7_ASTIGMATISM	Astigmatism	211 588	16 380 455	FinnGen
ukb-b-19994	Spherical power	99 380	9 851 867	UK Biobank
ukb-b-10250	Cylindrical power	99 380	9 851 867	UK Biobank
ukb-b-11480	CH	97 653	9 851 867	UK Biobank
ukb-b-15178	CRF	97 653	9 851 867	UK Biobank
ukb-b-19277	IOPcc	97 653	9 851 867	UK Biobank
ukb-b-14146	IOPg	97 653	9 851 867	UK Biobank

CH = corneal hysteresis; CRF = corneal resistance factor; IOPcc = corneal-compensated intraocular pressure; IOPg = Goldmann-correlated intraocular pressure; SNPs = single nucleotide polymorphisms.

with a P value >0.05 indicating the absence of heterogeneity. A regression intercept test was employed to detect horizontal pleiotropy, with an intercept value close to 0 indicating minimal pleiotropy. In the absence of heterogeneity and pleiotropy, the fixed-effect IVW method was employed. If heterogeneity was present but pleiotropy was absent, the random-effect IVW method was used. In cases of pleiotropy, regardless of heterogeneity, MR-Egger regression was implemented. Influential SNPs were identified using leave-one-out analysis.

A meta-analysis of the MR estimates derived from various GWAS data sources was conducted using the metafor package. Heterogeneity in the meta-analysis was evaluated using the I^2 statistic, with I^2 values of 25%, 50%, and 75% indicating low, medium, and high heterogeneity, respectively. The decision to use a fixed-effect model versus a random-effects model in our meta-analysis was based on 2 key considerations: (1) the results of the heterogeneity test and, more importantly, (2) whether all results share a common effect size. The same MR methodology was employed for the reverse MR analysis to investigate the causal relationship between refractive error and corneal biomechanical parameters.

Results

Instrumental Variables

In this study, the IVs were extensively checked for outliers and confounding factors. After removal of outliers and palindromic sequences, the mean F values exceeded 10, indicating that the selected IVs were highly reliable. For the corneal biomechanical parameters (CH, CRF, IOPcc, and IOPg), the numbers of effective SNPs were 109, 145, 53, and 93, respectively, with corresponding mean F values of 69.49, 71.35, 56.51, and 56.53, respectively. Further details on the associated SNPs are provided in Tables S2 to S5 (available at www.ophthalmologyscience.org).

Causal Effect of CH on Refractive Error

The analysis of myopia utilizing the UK Biobank database indicates that higher CH is causally associated with a lower likelihood of myopia development, with an OR of 0.9936 (95% CI = 0.9899−0.9973; $P = 7.79 \times 10^{-4}$) (Fig 2, Table S6, available at www.ophthalmologyscience.org). The causal associations are visually represented in scatter plots (Fig 3), and the results of the leave-one-out analysis indicate that the findings from the IVW analysis are not affected by any single SNP (Fig S4, available at www.ophthalmologyscience.org). MR-Egger regression was then used to evaluate the potential for average horizontal pleiotropy among the IVs, and funnel plots were utilized to visualize directional pleiotropy. If the MR-Egger intercept is not statistically significant ($P > 0.05$), the relative symmetry of the funnel plots (Fig S5, available at www.ophthalmologyscience.org) can be interpreted as an indicator of the absence of horizontal pleiotropy. The heterogeneity and horizontal pleiotropy results are presented in Table S7 (available at www.ophthalmologyscience.org). By contrast, the FinnGen consortium analysis did not identify a significant causal relationship between CH and myopia. However, a meta-analysis combining the results from both databases provides evidence of a causal link, yielding a combined OR of 0.9936 (95% CI = 0.9899−0.9973; $P = 0.0008$). Furthermore, the heterogeneity analysis revealed $I^2 = 0.00\%$, Q = 1.68, and $P = 0.43$, indicating that the combined OR was not significantly affected by heterogeneity across studies.

With regard to hyperopia, no significant causal relationship was found among the different populations, and the results of the meta-analysis did not indicate a causal relationship. Regarding the influence of CH on astigmatism, analysis of the UK Biobank database suggested that CH had a protective effect, with an OR of 0.9975 (95% CI = 0.9954−0.9997; $P = 0.02$). However, an analysis conducted by the FinnGen consortium did not reveal a significant causal relationship between CH and astigmatism. The results of the meta-analysis confirmed a causal relationship, with a combined OR of 0.9017 (95% CI = 0.6935−1.1724; $P = 0.440$), and heterogeneity analysis showed the following: $I^2 = 72.49\%$; Q = 3.64; and $P = 0.056$.

In terms of spherical power, CH is associated with an increase in spherical power, which correlates with a decreased likelihood of myopia development ($\beta = 0.0885$, 95% CI = 0.0410−0.1361; $P = 2.62 \times 10^{-4}$; Fig 6). In contrast, CH is associated with a reduction in cylindrical power ($\beta = -0.0287$, 95% CI = −0.056 to −0.0013; $P = 0.04$).

Causal Effect of CRF on Refractive Error

Analysis of the UK Biobank database indicated that higher CRF is causally associated with a lower likelihood of development, with an OR of 0.9946 (95% CI = 0.9912−0.9981; $P = 2.41 \times 10^{-3}$, Table S8, available at www.ophthalmologyscience.org). Conversely, an analysis conducted using the FinnGen consortium did not identify a causal relationship between CRF and myopia. The meta-analysis results corroborated the causal relationship between CRF and myopia, yielding a combined OR value of 0.9946 (95% CI = 0.9912−0.9981; $P = 0.0024$), with heterogeneity analysis revealing the following: $I^2 = 0.00\%$; Q = 0.0058; and $P = 0.94$.

In the context of hyperopia, no significant causal relationship was detected across the various databases, and the meta-analysis results did not indicate any causal association. With respect to astigmatism, the UK Biobank analysis shows that higher CRF significantly impacts the development of astigmatism, with an OR of 0.9977 (95% CI = 0.9959−0.9996; $P = 0.017$). However, the FinnGen consortium analysis did not reveal a significant causal relationship between CRF and astigmatism. The results of the meta-analysis further supported the causal relationship between CRF and astigmatism, yielding a combined OR of 0.9186 (95% CI = 0.7375−1.1442; $P = 0.4487$), with heterogeneity analysis indicating $I^2 = 70.45\%$, Q = 3.38, and $P = 0.0658$. Regarding spherical power, CRF was associated with increased spherical power, decreasing the likelihood of myopia development, as indicated by a β of 0.0689 (95% CI = 0.0374−0.1003; $P = 1.75 \times 10^{-5}$). No causal relationship was identified between the CRF and cylindrical power.

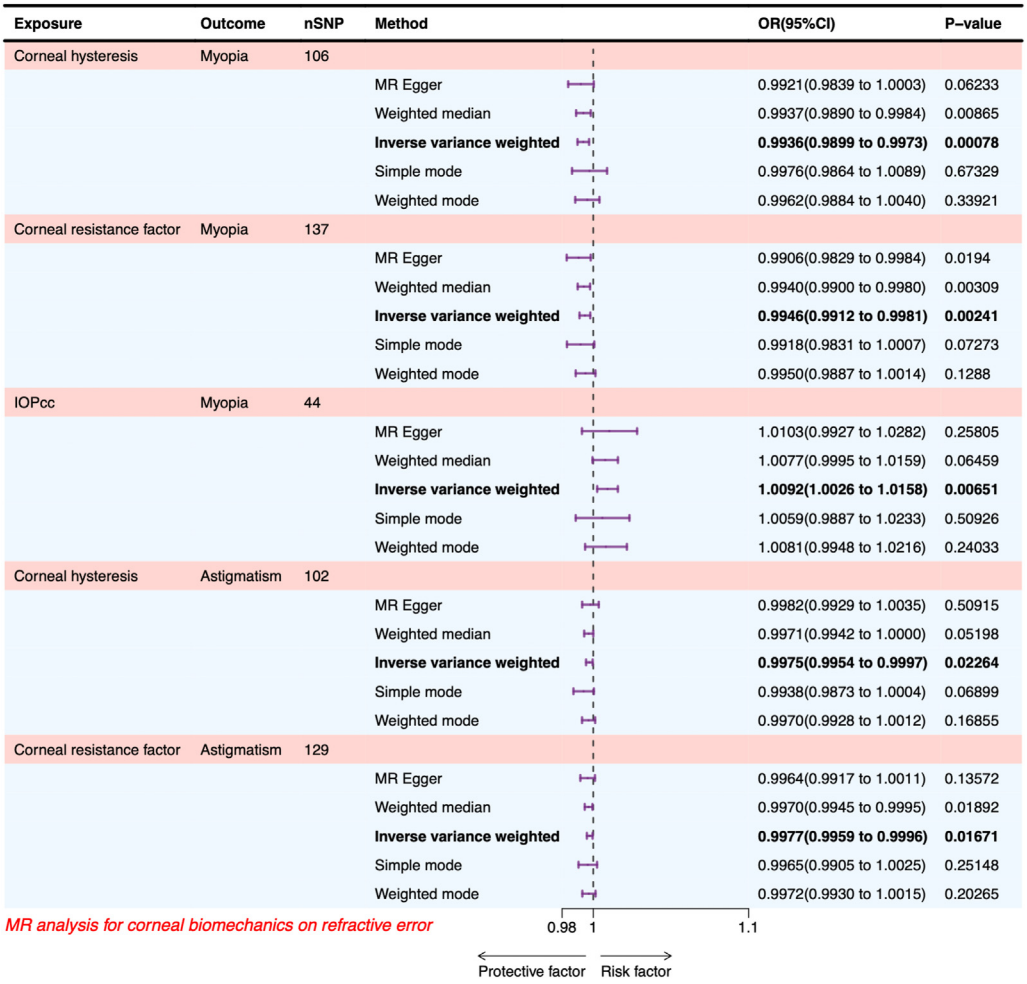


Figure 2. The figure shows the results of evaluating the causal effect of corneal biomechanical parameters on refractive errors (myopia and astigmatism) using 5 different methods of MR analysis (MR-Egger, weighted median, inverse variance weighting, simple mode, and weighted mode). The forest plot in the figure shows the ORs and their 95% CIs calculated by each method, with statistical significance determined by *P* values. Odds ratio values >1 are shown as risk factors, whereas OR values <1 are shown as protective factors. CI = confidence interval; IOPcc = corneal-compensated intraocular pressure; MR = Mendelian randomization; OR = odds ratio.

Causal Effect of IOPcc on Refractive Error

The results of MR indicated that the higher IOPcc variable from the UK Biobank database exhibited a positive causal relationship with myopia (OR = 1.0091, 95% CI = 1.0014−1.0169; $P = 2.07 \times 10^{-2}$, Table S9, available at www.ophthalmologyscience.org), whereas the IOPcc variable from the FinnGen consortium database did not demonstrate such a relationship. A meta-analysis combining the results from both databases supported a causal relationship, yielding a combined OR of 1.1185 (95% CI = 0.8428−1.4844; $P = 0.4379$). Heterogeneity analysis revealed an I^2 statistic of 67.31%, with a Q statistic of 3.06 and a *P* value of 0.0803. In contrast, no significant causal relationships were identified across the various databases for hyperopia and astigmatism, and the meta-analysis results did not indicate any causal associations. With respect to spherical power, the IOPcc variable demonstrated a negative causal effect, suggesting an increased likelihood of myopia ($\beta = -0.1653$, 95% CI = −0.2181 to −0.1124, $P = 8.83 \times 10^{-10}$). However, no causal relationship was observed between IOPcc and cylindrical power.

Causal Effect of IOPg on Refractive Error

There was no significant causal relationship between IOPg and refractive errors, including myopia, hyperopia, and astigmatism, across various databases (Table S10, available at www.ophthalmologyscience.org). Furthermore, the meta-analysis did not reveal any causal associations. Specifically, no significant causal relationship was identified between IOPg and spherical or cylindrical power.

Reverse MR Analysis

In the reverse MR analysis, refractive errors, specifically myopia, hyperopia, and astigmatism, and refractive power, including spherical and cylindrical power, were used as

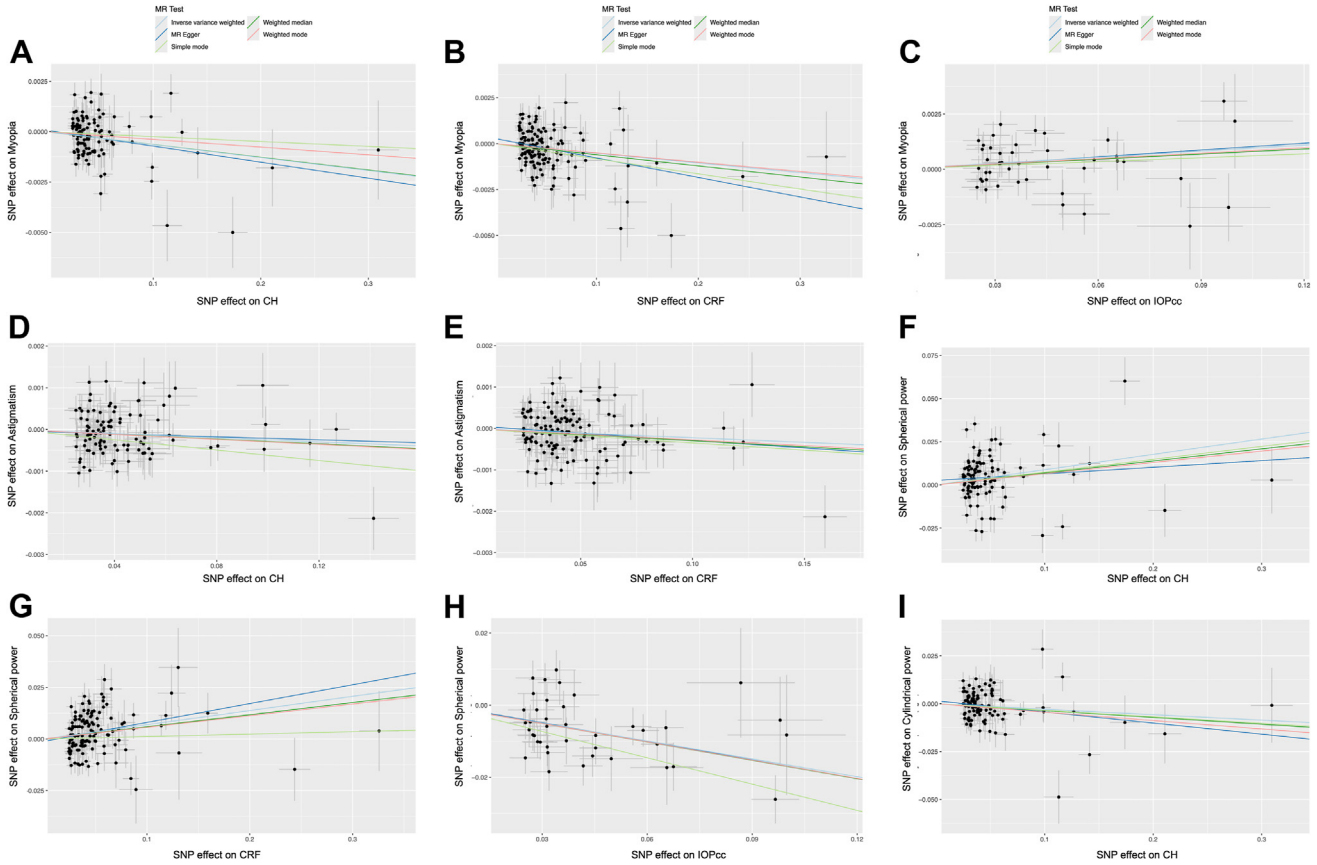


Figure 3. Scatterplots comparing the effects of SNPs on different parameters using 5 MR methods (inverse variance weighting, MR-Egger, simple mode, weighted median, and weighted mode). Each subplot reflects the relationship between SNP effects on the horizontal axis (corneal biomechanical parameters) and refractive status on the vertical axis. Black dots represent specific SNP effect estimates, and gray bands indicate the uncertainty of these estimates. The results of linear regression for the 5 methods are shown by lines of different colors. **A,** Single nucleotide polymorphism effects on CH and their association with myopia. **B,** Single nucleotide polymorphism effects on CRF and their association with myopia. **C,** Single nucleotide polymorphism effects on IOPcc and their association with myopia. **D,** Single nucleotide polymorphism effects on CH and their association with astigmatism. **E,** Single nucleotide polymorphism effects on CRF and their association with astigmatism. **F,** Single nucleotide polymorphism effects on CH and their association with spherical power. **G,** Single nucleotide polymorphism effects on CRF and their association with spherical power. **H,** Single nucleotide polymorphism effects on IOPcc and their association with spherical power. **I,** Single nucleotide polymorphism effects on CH and their association with cylindrical power. CH = corneal hysteresis; CRF = corneal resistance factor; IOPcc = corneal-compensated intraocular pressure; MR = Mendelian randomization; SNPs = single nucleotide polymorphisms.

exposure variables. Corneal biomechanical parameters, specifically CH and CRF, were used as the outcome variables. The results of the study indicate that, apart from spherical power as exposure and CH as outcome, which have a positive causal relationship ($\beta = 0.0643$, 95% CI = 0.0335−0.0951, $P = 4.32 \times 10^{-5}$) (Fig 7), none of the other parameters had a causal relationship with corneal biomechanical parameters (Table S11, available at www.ophthalmologyscience.org).

Discussion

This study employed MR analysis to investigate the causal relationships between corneal biomechanical parameters and various refractive errors, including myopia, hyperopia, and astigmatism, and refractive power, which encompasses spherical and cylindrical power. The meta-analysis incorporated data from 2 large-scale consortia: UK Biobank and FinnGen. These findings indicate that corneal biomechanical parameters exert differential causal effects on different refractive states. Specifically, CH and CRF were protective against the development of myopia. Conversely, IOPcc potentially contributed to the development of myopia.

The ORA is used to measure CH and CRF, which are crucial indicators of corneal biomechanical properties.[17] These measurements provide valuable insights into the corneal response to external stress and its stability. Corneal hysteresis primarily reflects the viscous properties of the cornea, indicating its ability to absorb and cushion impact or pressure.[13] In contrast, the CRF is associated with the overall stiffness of the cornea, representing its resistance to deformation.[13] Both these indicators are critical for maintaining the structural integrity and functional performance of the cornea. Both CH and CRF

Exposure	Outcome	nSNP	Method	β(95%CI)	P–value
Corneal hysteresis	Spherical power	106			
			MR Egger	0.0385(−0.0663 to 0.1434)	0.47304
			Weighted median	0.0695(0.0280 to 0.1110)	0.00102
			Inverse variance weighted	**0.0885(0.0410 to 0.1361)**	**0.00026**
			Simple mode	0.0743(−0.0313 to 0.1799)	0.17106
			Weighted mode	0.0651(0.0038 to 0.1264)	0.03993
Corneal resistance factor	Spherical power	127			
			MR Egger	0.0914(0.0184 to 0.1644)	0.01545
			Weighted median	0.0588(0.0236 to 0.0940)	0.00107
			Inverse variance weighted	**0.0689(0.0374 to 0.1003)**	**2e−05**
			Simple mode	0.0117(−0.0762 to 0.0996)	0.79461
			Weighted mode	0.0563(0.0092 to 0.1035)	0.02068
IOPcc	Spherical power	40			
			MR Egger	−0.1717(−0.3154 to −0.0280)	0.02449
			Weighted median	−0.1699(−0.2318 to −0.1081)	7.15e−08
			Inverse variance weighted	**−0.1653(−0.2181 to −0.1124)**	**8.83e−10**
			Simple mode	−0.2430(−0.3630 to −0.1230)	3e−04
			Weighted mode	−0.1692(−0.2577 to −0.0807)	0.00058
Corneal hysteresis	Cylindrical power	106			
			MR Egger	−0.0581(−0.1184 to 0.0022)	0.06181
			Weighted median	−0.0358(−0.0745 to 0.0029)	0.06994
			Inverse variance weighted	**−0.0287(−0.0560 to −0.0013)**	**0.04008**
			Simple mode	−0.0342(−0.1141 to 0.0458)	0.40406
			Weighted mode	−0.0441(−0.0903 to 0.0022)	0.06476

MR analysis for corneal biomechanics on refractive power

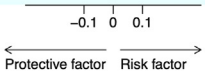


Figure 6. The figure shows the results of evaluating the causal effect of corneal biomechanical parameters on refractive power (spherical and cylindrical power) using 5 different methods of MR analysis. The forest plot in the figure shows the beta coefficients (β) and their 95% CIs calculated by each method, with statistical significance determined by *P* values. β value >0 are shown as risk factors, whereas β value <0 are shown as protective factors. CI = confidence interval; MR = Mendelian randomization.

are obtained from the ORA, and these parameters are currently the only in vivo corneal biomechanical data available in the GWAS database. Variations in these values may also influence the development and progression of refractive errors.[18]

The relationship between corneal biomechanics and refractive errors has been debated in previous studies. Most studies have indicated that CH and CRF are significantly reduced in patients with myopia,[19–22] particularly in those with high myopia.[9,23] This reduction suggests a compromise

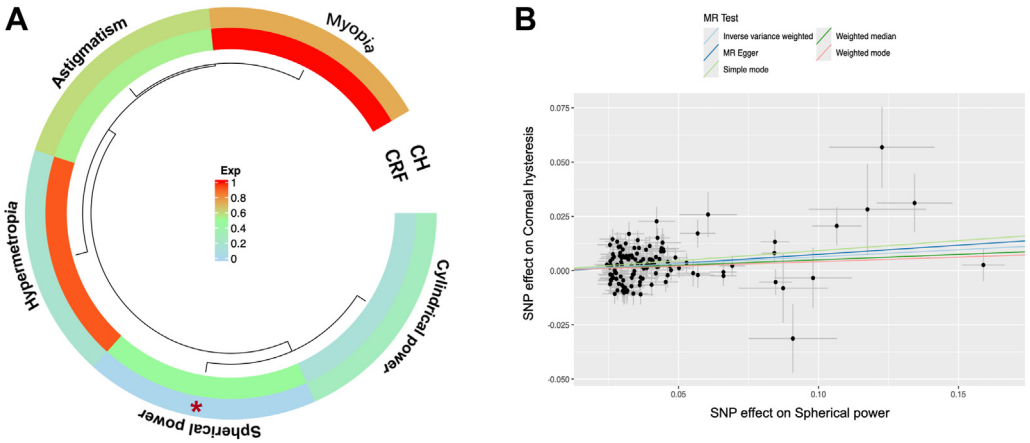


Figure 7. Causal analysis of CH and CRF on refractive status based on reverse MR analyses. **A,** Circular heatmap of MR IVW results among different traits. The *P* values of CH and CRF are shown from the outside to the inside. An asterisk indicates that the MR result is causal ($P < 0.05$). **B,** Scatter plots showing the SNP effects on spherical power and their association with CH. CH = corneal hysteresis; CRF = corneal resistance factor; IVW = inverse variance weighted; MR = Mendelian randomization; SNPs = single nucleotide polymorphisms.

between the corneal elasticity and resistance, which increases the risk of ocular deformation. A meta-analysis by Wu et al[9] provided substantial support for this assertion. This decrease may be linked to the interaction between collagen fibrils and the extracellular matrix in the corneal stroma, reflecting a decline in corneal viscoelasticity and mechanical strength.[24] This phenomenon has been confirmed in adults[6] and young children.[25] Collectively, these findings imply that changes in the corneal biomechanics may contribute to the pathological processes associated with myopia. Subsequent investigations by Han et al[5] and Jiang et al[22] further supported this view, showing a positive correlation between corneal biomechanical parameters and the spherical equivalent. Moreover, a study by Long et al[26] on preschool children in China demonstrated that the corneal biomechanical characteristics of myopic eyes tend to weaken at an early developmental stage, suggesting that alterations in corneal biomechanics are an early initiating factor in myopia development. Consequently, these studies support the hypothesis that corneal biomechanics may serve as an underlying cause of myopia rather than merely a consequence of its progression.

Conversely, some studies have reported no significant correlation between CH and CRF with myopia progression.[27,28] This finding adds an additional layer of complexity and controversy to this area of research. These discrepancies in the results may be attributed to the varying degrees and types of myopia present in the study populations, underscoring the need for a more consistent sample selection and classification in future research. Furthermore, previous studies predominantly utilized cross-sectional datasets, which inherently limit the ability to establish causal relationships, affecting the reliability of their conclusions. Although some studies have employed a 2-year follow-up to mitigate these concerns, the quality of ORA measurements obtained from children remains a significant intrinsic limitation.[25] Such inaccuracies may obscure or artificially inflate the perceived relationship between corneal biomechanics and myopia.

To further investigate the causal relationship between CH, CRF, and the development of myopia, this study employed the MR method. This approach employs genetic variation as a natural experiment to effectively eliminate the influence of potential confounders, providing robust support for causal inference.[29] UK Biobank data revealed significant causal relationships between CH, CRF, and the development of myopia. However, similar validation was not achieved in the FinnGen consortium. Nevertheless, the meta-analysis results indicated the presence of a causal relationship. This discrepancy in the findings may be due to differences in sample characteristics, environmental factors, and genetic backgrounds between the 2 databases. These findings suggest that the corneal biomechanical properties exert a regulatory influence on myopia.

Although CH and CRF have been shown to protect against the development of myopia, IOPcc tends to promote myopia development. This finding supports the hypothesis that IOPcc affects ocular mechanics, altering the axial length and affecting the refractive status.[30,31] However, no significant causal effects were found for hyperopia or astigmatism, suggesting that IOPcc has a more pronounced effect on myopia development than other refractive states. In contrast, IOPg did not demonstrate significant causal effects on refractive error. The different results for IOPg and IOPcc suggest that these 2 IOP measurements reflect distinct biomechanical characteristics.[32] Goldmann-correlated intraocular pressure is calibrated against the Goldmann applanation tonometer, whereas IOPcc is adjusted for corneal biomechanical properties, considering factors such as corneal thickness and elasticity.[32] Central corneal thickness is associated with IOPg but not with IOPcc.[19,32] Therefore, IOPcc reflects the overall structure and deformation characteristics of the eye more accurately. Changes in corneal biomechanical properties, such as reduced elasticity and resistance to deformation, are closely associated with myopia development.[31] This corneal biomechanical adjustment makes IOPcc more sensitive to changes in ocular structure, explaining its positive correlation with myopia. In contrast, IOPg does not fully account for corneal biomechanical factors, limiting its association with myopia and its ability to capture the potential impact of corneal biomechanical changes.

Although most current research focuses on myopia, the relationship between corneal biomechanical parameters and astigmatism should not be ignored. The results of this study confirm that CH and CRF play causal roles in suppressing the development of astigmatism. The elasticity and resistance of the cornea are important for maintaining its regular shape, and the stability of these shapes is closely related to the development of astigmatism. In eyes with astigmatism, the cornea may exhibit irregularities because of insufficient elasticity, which may contribute to the development of astigmatism. Therefore, corneal biomechanical properties play an important role in maintaining corneal symmetry and shape stability.

With regard to hyperopia, the available literature suggests that CH and CRF increase in individuals with hyperopia,[33] with increased refractive power resulting in a notable enhancement in corneal elasticity and resistance. İnceoğlu et al[31] observed a significant increase in CH from high myopia to high hyperopia, which may indicate that the cornea of patients with hyperopia exhibits greater stiffness and thus greater resistance to deformation. However, the present study did not establish a causal relationship between CH and CRF in hyperopia. Similarly, Long et al[26] observed in a study of preschool children in China that the corneal biomechanical properties in hyperopic eyes did not undergo significant alterations, which may be associated with the developmental characteristics of the cornea in children. This may be because hyperopia is primarily influenced by axial length and corneal curvature rather than directly by corneal biomechanical properties. Furthermore, the literature indicates that flatter corneas are typically associated with lower CH and CRF readings. This may explain the weak association between hyperopia and corneal biomechanics.

The results of reverse MR analysis demonstrated that only spherical power had a causal effect on CH. This may be because myopia causes an increase in the axial length of the

eye rather than myopia itself causing a reduction in the corneal resistance to deformation. This also explains why some studies have suggested that only high myopia is associated with mechanical properties, whereas moderate and low myopia show no significant correlations. Corneal biomechanical properties may play a significant role in the development of myopia; however, the effect of myopia on corneal biomechanical properties seems to be minimal or insignificant. The relationship between refraction and corneal biomechanical parameters is complex, and more in-depth studies are required to explore the mechanisms of their interaction.

One limitation of our study is that, although our MR analysis reveals significant associations between corneal biomechanics and myopia, the effect sizes are relatively small. Nevertheless, these modest changes could still hold clinical significance, particularly for certain high-risk individuals. Furthermore, the small effect sizes may exert a cumulative impact at the population level. While the individual risk changes may seem minimal, their overall effect in a large population could be substantial. Future research should focus on exploring the underlying biological mechanisms that connect corneal biomechanics to refractive errors. Additionally, conducting larger, multicenter studies to validate these findings across diverse populations will be essential.

In conclusion, corneal biomechanical properties may protect against refractive errors like myopia. Clinical use of these parameters could aid prediction, but further studies are needed to confirm causality and explore mechanisms using advanced imaging tools (e.g., Corvis ST).

Footnotes and Disclosures

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Data collection: Wei, Su, Jia

Analysis and interpretation: Wei, Han, Wang

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No animal subjects were used in this study.

Abbreviations and Acronyms:
CI = confidence interval; **CH** = corneal hysteresis; **CRF** = corneal resistance factor; **GWAS** = genome-wide association study; **IOPcc** = corneal-compensated intraocular pressure; **IOPg** = Goldmann-correlated intraocular pressure; **IVs** = instrumental variables; **IVW** = inverse variance weighted; **MR** = Mendelian randomization; **OR** = odds ratio; **ORA** = ocular response analyzer; **SNPs** = single nucleotide polymorphisms.

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